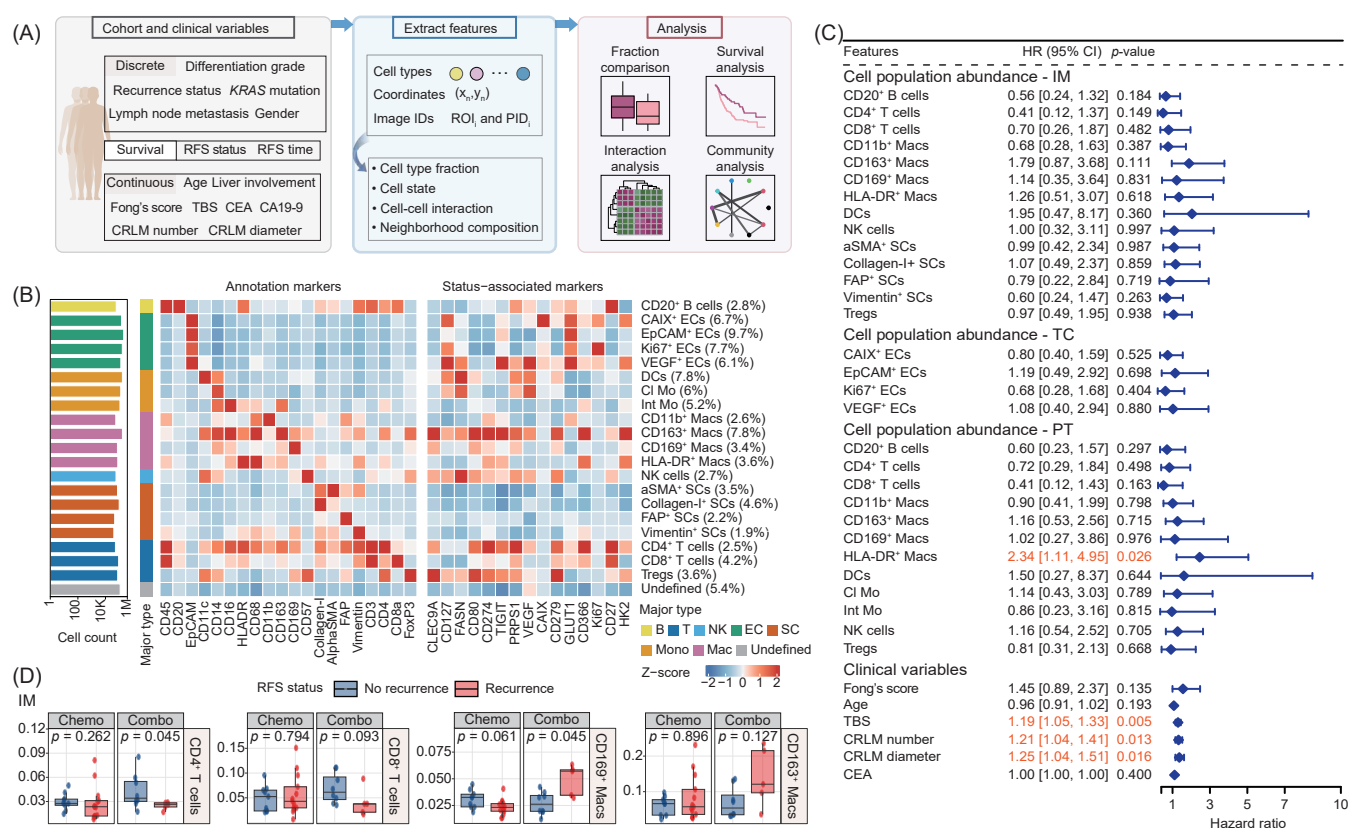

(A)
Cohort and clinical variables
Discrete
Differentiation grade
Recurrence status
KRAS mutation
Lymph node metastasis
Gender
Survival
RFS status
RFS time
Continuous
Age
Liver involvement
Fong's score
TBS
CEA
CA19-9
CRLM number
CRLM diameter
Extract features
Cell types
Coordinates
Image IDs
ROIi and PIDi
• Cell type fraction
• Cell state
• Cell-cell interaction
• Neighborhood composition
Analysis
Fraction comparison
Survival analysis
Interaction analysis
Community analysis
(B)
Annotation markers
Status−associated markers
CD20+ B cells (2.8%)
CAIX+ ECs (6.7%)
EpCAM+ ECs (9.7%)
Ki67+ ECs (7.7%)
VEGF+ ECs (6.1%)
DCs (7.8%)
Cl Mo (6%)
Int Mo (5.2%)
CD11b+ Macs (2.6%)
CD163+ Macs (7.8%)
CD169+ Macs (3.4%)
HLA-DR+ Macs (3.6%)
NK cells (2.7%)
aSMA+ SCs (3.5%)
Collagen-I+ SCs (4.6%)
FAP+ SCs (2.2%)
Vimentin+ SCs (1.9%)
CD4+ T cells (2.5%)
CD8+ T cells (4.2%)
Tregs (3.6%)
Undefined (5.4%)
Cell count
Major type
CD45 CD20 EpCAM CD11c CD14 CD16 HLADR CD68 CD11b CD163 CD169 CD57 Collagen-I AlphaSMA FAP Vimentin CD3 CD4 CD8a FoxP3
CLEC9A CD127 FASN CD80 CD274 TIGIT PRPS1 VEGF CAIX CD279 GLUT1 CD366 Ki67 CD27 HK2
Major type
B T NK EC SC
Mono Mac Undefined
Z−score
−2 −1 0 1 2
(C)
Features
HR (95% CI)
p-value
Cell population abundance - IM
CD20+ B cells 0.56 [0.24, 1.32] 0.184
CD4+ T cells 0.41 [0.12, 1.37] 0.149
CD8+ T cells 0.70 [0.26, 1.87] 0.482
CD11b+ Macs 0.68 [0.28, 1.63] 0.387
CD163+ Macs 1.79 [0.87, 3.68] 0.111
CD169+ Macs 1.14 [0.35, 3.64] 0.831
HLA-DR+ Macs 1.26 [0.51, 3.07] 0.618
DCs 1.95 [0.47, 8.17] 0.360
NK cells 1.00 [0.32, 3.11] 0.997
aSMA+ SCs 0.99 [0.42, 2.34] 0.987
Collagen-I+ SCs 1.07 [0.49, 2.37] 0.859
FAP+ SCs 0.79 [0.22, 2.84] 0.719
Vimentin+ SCs 0.60 [0.24, 1.47] 0.263
Tregs 0.97 [0.49, 1.95] 0.938
Cell population abundance - TC
CAIX+ ECs 0.80 [0.40, 1.59] 0.525
EpCAM+ ECs 1.19 [0.49, 2.92] 0.698
Ki67+ ECs 0.68 [0.28, 1.68] 0.404
VEGF+ ECs 1.08 [0.40, 2.94] 0.880
Cell population abundance - PT
CD20+ B cells 0.60 [0.23, 1.57] 0.297
CD4+ T cells 0.72 [0.29, 1.84] 0.498
CD8+ T cells 0.41 [0.12, 1.43] 0.163
CD11b+ Macs 0.90 [0.41, 1.99] 0.798
CD163+ Macs 1.16 [0.53, 2.56] 0.715
CD169+ Macs 1.02 [0.27, 3.86] 0.976
HLA-DR+ Macs 2.34 [1.11, 4.95] 0.026
DCs 1.50 [0.27, 8.37] 0.644
Cl Mo 1.14 [0.43, 3.03] 0.789
Int Mo 0.86 [0.23, 3.16] 0.815
NK cells 1.16 [0.54, 2.52] 0.705
Tregs 0.81 [0.31, 2.13] 0.668
Clinical variables
Fong's score 1.45 [0.89, 2.37] 0.135
Age 0.96 [0.91, 1.02] 0.193
TBS 1.19 [1.05, 1.33] 0.005
CRLM number 1.21 [1.04, 1.41] 0.013
CRLM diameter 1.25 [1.04, 1.51] 0.016
CEA 1.00 [1.00, 1.00] 0.400
1 3 5 7 10
Hazard ratio
(D)
IM
RFS status
No recurrence
Recurrence
Chemo
Combo
p = 0.262
p = 0.045
CD4+ T cells
p = 0.794
p = 0.093
CD8+ T cells
p = 0.061
p = 0.045
CD169+ Macs
p = 0.896
p = 0.127
CD163+ Macs